

A4

September 25, 2025

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[17]: import numpy as np
import matplotlib.pyplot as plt

x = [0.54569285, 0.66713963, 1.8821619, 2.62643799, 2.77755533, 0.62036703, 2.
↳ 19165391, 2.23670046, 2.8954384, 1.46367319, 2.89139908, 1.28584076, 0.
↳ 91031887, 1.58378549, 2.70337433, 1.06220882, 0.62533654, 0.2543899, 1.
↳ 84942398, 1.02288247, 1.32818655, 0.77015435, 1.32683635, 2.14699776, 1.
↳ 72578687, 1.07192398, 1.41602911, 1.45684425, 0.93492731, 2.17288046, 2.
↳ 38366803, 0.18553194, 1.16055873, 1.91454921, 0.67359843, 1.96104196, 2.
↳ 05537434, 0.73697772, 1.67134223, 1.27495446, 2.05094272, 2.7166306, 1.
↳ 67999956, 2.90449579, 0.47185534, 2.30635976, 2.34910576, 2.52596201, 1.
↳ 81093635, 0.61504781, 2.25432395, 2.58362244, 1.32451323, 2.66484963, 2.
↳ 28977926, 1.59435303, 0.16220664, 2.74076023, 2.13411313, 0.24886403, 2.
↳ 75937128, 2.52511043, 0.79023342, 0.8626656, 2.57910037, 1.4577751, 0.
↳ 69469835, 1.1755677, 2.30071142, 1.31623152, 1.7563655, 2.42753782, 1.
↳ 63616482, 2.07380807, 0.27889959, 1.26687733, 0.19751214, 1.7340717, 0.
↳ 48812455, 1.46163183, 1.05348728, 1.02458033, 0.33707579, 2.34180852, 0.
↳ 4797847, 1.75803667, 2.33557441, 1.22958469, 1.64746791, 1.91579049, 0.
↳ 93783358, 1.73717707, 0.46471257, 0.64567871, 0.72231835, 0.2588409, 2.
↳ 48499575, 1.95993974, 2.10266381, 1.33524309]

y = [0.53514393, 0.58600211, 0.93310423, 0.52689839, 0.40936582, 0.50996668, 0.
↳ 73574805, 0.7192478, 0.28952536, 1.04168546, 0.09719999, 1.09654248, 0.
↳ 85966593, 1.24210298, 0.14967488, 0.81268619, 0.62098107, 0.27431869, 0.
↳ 66311695, 0.81421226, 0.91617078, 0.8182873, 1.10528536, 0.82232858, 1.
↳ 01361609, 0.81097813, 0.94504491, 0.91921348, 0.88720014, 0.68272353, 0.
↳ 78465141, 0.18418327, 0.89964478, 0.90807661, 0.5551873, 0.71194505, 0.
↳ 86465198, 0.48713479, 1.01252095, 0.9007005, 0.86324039, 0.51321692, 0.
↳ 8931645, 0.28491966, 0.46861117, 0.80306315, 0.52843247, 0.56275205, 0.
↳ 9746029, 0.5096404, 0.91763387, 0.44502913, 1.09550101, 0.44277292, 0.
↳ 74395972, 0.97298009, 0.24682351, 0.34909826, 0.58169838, 0.21565899, 0.
↳ 37707359, 0.55114338, 0.62163084, 0.79015231, 0.67503296, 1.25182691, 0.
↳ 76448087, 1.18548521, 0.67909665, 0.98799479, 0.88643866, 0.71101252, 0.
↳ 95385419, 0.909671, 0.31834095, 1.01003781, 0.21067932, 0.85834788, 0.
↳ 4005094, 0.91000894, 0.66392828, 0.92155107, 0.31528422, 0.77240702, 0.
↳ 62354821, 0.89624496, 0.70959879, 0.90442675, 1.14458026, 0.87690613, 0.
↳ 99032025, 1.08341817, 0.53103793, 0.58917849, 0.67510364, 0.16816688, 0.
↳ 48439669, 1.08512395, 0.93636791, 1.07801109]
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polregdeg1 = np.poly1d(np.polyfit(x, y, 1))
polregdeg2 = np.poly1d(np.polyfit(x, y, 2))
polregdeg3 = np.poly1d(np.polyfit(x, y, 3))
plot = np.linspace(min(x), max(x), 100)
plt.scatter(x, y)

plt.plot(plot, polregdeg1(plot), label = 'Deg 1')
plt.plot(plot, polregdeg2(plot), label = 'Deg 2')
plt.plot(plot, polregdeg3(plot), label = 'Deg 3')

plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Degree 1, 2, and 3 Polynomial Regressions')
plt.legend()
plt.show()

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